Fiscal Year:	FY 2020	**Task Last Updated:**	FY 10/29/2020
PI Name:	Lockley, Steven W Ph.D.		
Project Title:	Development and Testing of Biomarkers to Determine Individual Astronaut Vulnerabilities to Behavioral Health Disruptions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	slockley@hms.harvard.edu	**Fax:**	FY 617 732-4015
PI Organization Type:	UNIVERSITY	**Phone:**	617-732-4977
Organization Name:	Brigham and Women's Hospital		
PI Address 1:	Division of Sleep Medicine and Circadian Disorders		
PI Address 2:	221 Longwood Avenue, Suite 438		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	08/01/2014	**End Date:**	01/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Williams, Thomas	**Contact Phone:**	281-483-8773
Contact Email:	thomas.j.will1@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 1/31/2020 per NSSC information (Ed., 9/13/19) NOTE: Extended to 7/31/2019 per NSSC information (Ed., 8/7/18) NOTE: Extended to 7/31/2018 per K. Ohnesorge/JSC (Ed., 7/14/17) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/18/17) NOTE: End date shows as 7/31/2017 per NSSC information (Ed., 6/2/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Barger, Laura Ph.D. (Brigham & Women's Hospital) Czeisler, Charles M.D., Ph.D. (Brigham and Women's Hospital/Harvard Medical Center) Duffy, Jeanne Ph.D. (Brigham & Women's Hospital) Kristal, Bruce Ph.D. (Brigham & Women's Hospital)		
Grant/Contract No.:	NNX14AK53G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	It is well established that sleep deficiency and circadian desynchrony impair alertness and performance. It is also increasingly recognized that the response to sleep loss and circadian desynchrony is highly individualized and both challenge- and task-dependent. This high degree of inter-individual variability makes it very difficult to predict who will be at risk and when they will be at risk, and therefore a 'one-size-fits-all' approach is likely to be suboptimal. What is needed are biomarkers of individual vulnerabilities and resiliencies to sleep loss and circadian desynchronization that can be used to predict problems and intervene when necessary. In Phase 1 of this project, we conducted a comprehensive and standardized analysis of individualized neurocognitive and psychological responses to sleep loss and circadian desynchrony from existing laboratory data in over 150 subjects and more than a thousand bed-days incorporating multiple measures of neurocognitive performance. These analyses have confirmed the best candidates for predicting neurocognitive impairment in simulated sleep and circadian challenges, common in space missions that are studied routinely in our laboratory. In Phase 2 of this project, we have conducted a series of short (6-day), targeted laboratory studies to test prospectively the predictive power of selected biomarkers and to measure novel biomarkers absent from our original data sets in order to inform operational analog studies testing the predictive value of these biomarkers in operational conditions. Our innovative goal is to generate sets of biomarker panels from lipidomic and/or metabolomic biomarker profiles to identify and predict interindividual differences in performance under acute sleep deprivation and circadian misalignment. In Phase 3 of this project, we aimed to conduct a pilot study testing the feasibility and utility of implementing the predictive biomarkers for neurocognitive impairment in Antarctica as a high fidelity space analog.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of a candidate biomarker panel that delineates inter-individual differences in sleep and circadian rhythms has enormous potential for the development of individualized predictive models of vulnerabilities and resiliencies to sleep deprivation and circadian misalignment for a wide range of occupational settings, for example, in shift-workers and other unusual environments (military, shipping, mining, etc).
	Phase 1. We have completed Phase 1 of the project, namely the development of a preliminary candidate biomarker panel that can predict an individual's response to sleep loss phenotype (i.e., resilient, intermediate, or vulnerable, as defined by performance at 20 hours of wakefulness) from neurobehavioral performance collected in the individual at baseline after a habitual (8 hour) night of sleep 2 days earlier. The results are published in St Hilaire et al., Scientific Reports, 2019 [see also Bibliography section]. Briefly, using a model based on lapses on the Psychomotor Vigilance Test (PVT), we found that we could discriminate ~50% of our subject population (studied under inpatient laboratory conditions) as non-resilient to sleep loss (i.e., intermediate or vulnerable response to sleep loss) at ~20 h awake from one neurobehavioral performance assessment during rested conditions (i.e., at ~8 hours of wake after a habitual 8-hour night of sleep) with ~100% accuracy. Our findings suggest that these individuals who are poor performers during rested conditions exhibit non-sleepiness-related performance impairment that is exacerbated by subsequent sleep loss. Among the remaining ~50% of our subject population, who are high-level performers during rested conditions, the model had an accuracy of only 67%, suggesting that neurobehavioral performance alone is insufficient to discriminate response to sleep loss phenotype in these individuals. Phase 2. We completed data collection for the laboratory study outlined for Phase 2 of this project. We consented and screened a total of 60 potential candidates. Forty-five were excluded due to not meeting study inclusion/exclusion criteria and we have completed studies on 15 individuals. Based on our Phase 1 analysis, participants were screened for performance prior to admission and excluded if they had 2 or more lapses on at least 2 screening test sessions. Among the 15 participants who completed the study, analysis of their performance data indicates that 11 participants were resilient, 3 participants were intermediate, and 1 participant was vulnerable at 20 hours awake during the 50-hour constant routine. Each participant was studied for 6 days in the laboratory. During days 1-3, participants completed baseline measures under stable well-rested conditions (8 h sleep per night) to provide individual control data for assessing the degree of impairment due to sleep loss or circadian desynchrony. On days 4-6, participants underwent 50 hours of acute sleep deprivation under constant routine conditions in order to assess inter-individual responses to acute sleep deprivation and being awake at an adverse circadian phase. Participants were allowed a final 10-hour recovery sleep before discharge. Throughout the 6-day study, we collected tests of performance and alertness every 1-2 hours while awake. Blood samples were taken at 60-minute intervals, and saliva was collected every hour during wake as backup for the plasma samples. Our initial analyses have focused on every 4th sample (every 4 hours) from Day 2 (to document the diurnal pattern of the biomarkers under well-rested conditions) through the end of the constant routine (CR). We completed lipidomics and metabolomics assays in the four-hourly samples in 12 participants (n=6F, 35.8 ± 8.1 yrs). Biomarker candidate characterization for circadian phase. Lipids. The number of unique lipid species that were identified in males and females were 335 and 351, respectively, which included 4 lipid classes. There was evidence of robust endogenous circadian regulation of the plasma lipidome with approximately 35% and 19% being rhythmic under ambulatory and constant routine conditions, respectively, in males and 24% and 27% in females. Triglycerides (TG) were the most abundant subclass of lipids that were rhythmic in both sexes under both behavioral states. Given that no lipids were circadian in all 12 participants, we lowered the threshold to include at least 50% of the population to being rhythmic for each lipid, and identified 29 unique lipids in the males and 76 in the females. Of these, only 19 unique lipids were circadian in both males and females. Taken together, these results suggest large inter-individual and sex-specific differences in the circadian regulation of the plasma lipidome. Identifying lipidomics biomarkers of circadian phase may require developing models that account for potential sex differences. Metabolites. The number of plasma metabolites that were chromatographically identified in males and females were 117 and 124, respectively. There was evidence of robust endogenous circadian regulation of plasma metabolites in both males and females. Approximately 36% and 25% were rhythmic under ambulatory and constant routine conditions,

Task Progress:	respectively, in males and 33% and 23%, respectively in females. Of these, only 6 unique metabolites were circadian in both males and females. As for lipids, there are large inter-individual and sex-specific differences in the circadian regulation of plasma metabolites. The majority (~60%) of the metabolites remained unchanged between baseline and constant routine regardless of whether they were rhythmic or not. Biomarker candidate characterization for time awake (sleepiness). Lipids. Similar to circadian regulation, there was evidence of extensive inter-individual variability in linear changes. In the males, approximately 5% and 32% were linear under baseline and constant routine conditions, respectively, with 6% and 34% for the females. The highest percentage of linear lipid profiles were seen in the TG subclass in both males and females, approximately 44%. The Lysophospholipid (LPC) subclass had a higher percent change in females (~45%) compared to males (~17%), highlights potential sex differences in the lipids that change linearly with time awake. Of the lipid features that changed linearly, approximately 50% of them increased in the males during the 16-h wake episode under ambulatory conditions versus ~70% increased in females. Under CR conditions, ~60% increased in males and ~67% in females. Metabolites. As in the lipids, the proportion of metabolites that were linear during baseline and CR conditions were similar for males (6% and 32%) and females (6% and 35%, respectively), although the individual candidates were not the same: Of the 80 metabolites that were linear between males and females during CR in at least 50% of the population, only 10 were shared by males and females. The direction of change was similar between males and females with ~70% decreasing during baseline and 55% decreasing during constant routine. Predictive biomarker models. Prediction of endogenous circadian phase and phase angle of entrainment. A preliminary predictive model was developed using the concentrations of lipid species that were (1) rhythmic under both ambulatory and CR conditions; (2) were rhythmic under CR in at least 2 participants; and (3) the phase of the lipid rhythm did not differ by more than 5 hours between individuals. The number of unique lipid species that met these criteria across all 12 participants was 52 and all were included in developing the predictive model. Partial Least Squares Regression (PLSR) yielded a predictive model that was able to use the lipid concentrations measured in a single blood sample collected ~4 h after waking on the day to predict endogenous circadian phase and phase angle of entrainment to be within, on average, ~20 and ~42 minutes of actual, respectively. While these analyses are preliminary, they are encouraging in that they demonstrate reasonable accuracy in predicting circadian phase, albeit the data are likely overfit. We are currently evaluating the model using additional time points, and fewer lipid features to further test the robustness of these predictions. Prediction of neurobehavioral performance impairment due to sleep loss. The number of unique lipid species common across all 12 participants initially included in the preliminary predictive model for sleepiness was 289. Five of the 12 participants were identified as vulnerable to sleep loss at 40 hours awake on CR; the remaining participants were identified as resilient. All participants were high-level performers during ambulatory baseline, which was defined as exhibiting only 0 or 1 lapse in attention on the PVT at 8 hours awake following an 8-hour time-in-bed opportunity. Supervised learning methods yielded predictive models that were able to correctly discriminate between resilient and vulnerable individuals using just 1-2 lipids measured at a single time point. Specifically, both k-nearest neighbor and naïve Bayes classifiers trained on lipidomic data at 8 hours awake on CR were able to discriminate resilient from vulnerable individuals with accuracies ranging from 83%-100% when tested on data collected at 8 hours awake two days earlier under ambulatory baseline. We are currently developing additional predictive models at additional time points, including predictions for less extreme acute sleep loss (e.g., 24 hours). We are also developing similar models using metabolomics data. Phase 3: We conducted similar analyses to those that we conducted in the Phase 1 analysis to identify whether baseline performance alone could accurately discriminate response to sleep loss among individuals overwintering in Antarctica. Unlike the laboratory data in which participants were kept awake for up to 50 hours, however, few individuals overwintering in Antarctica in our previously collected dataset reported wakefulness beyond 18 hours, and therefore our analysis was limited to predicting response to sleep loss >16 hours awake only. As was observed in the Phase 1 analysis, although baseline performance could identify high-level from poor performers at baseline, we could not further discriminate the response to sleep loss phenotype among high-level performers from baseline performance alone; we found that we were able to discriminate the response phenotype at >16 hours of wakefulness from baseline performance with an overall accuracy of ~64%.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Articles in Peer-reviewed Journals	St Hilaire MA, Kristal BS, Rahman SA, Sullivan JP, Quackenbush J, Duffy JF, Barger LK, Gooley JJ, Czeisler CA, Lockley SW. "Using a single daytime performance test to identify most individuals at high-risk for performance impairment during extended wake." Sci Rep. 2019 Nov 13;9(1):16681. https://doi.org/10.1038/s41598-019-52930-y ; PMID: 31723161; PMCID: PMC6853981 , Nov-2019
Articles in Peer-reviewed Journals	Kent BA, Rahman SA, St Hilaire MA, Grant LK, Rüger M, Czeisler CA, Lockley SW. "Circadian lipid and hepatic protein rhythms shift with a phase response curve different than melatonin." Nat Commun. 2022 Feb 3;13(1):681. https://doi.org/10.1038/s41467-022-28308-6 ; PMID: 35115537; PMCID: PMC8814172 , Feb-2022